Ch. 16 Solutions

Principles of Microeconomics

August 6, 2026

Quick Quiz

1. (c) – Without patents and copyright laws, firms and creators have less of an incentive to do research, invent new products, and create better art. Granting firms and creators temporary monopoly power has the potential to incentivize a more socially optimal amount of those activities.

2. (d) – If a firm exhibits decreasing average total cost over the range of relevant output, then it can produce all of the output for the market more cheaply than multiple firms could. As a result, other firms are unlikely to compete.

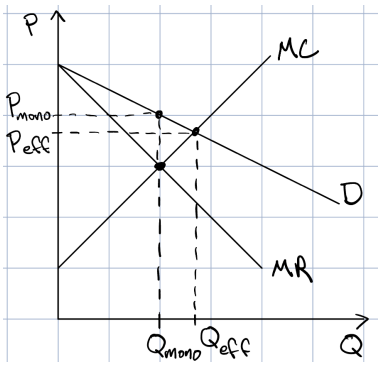
3. (b) – Marginal revenue equals marginal cost because monopolies are profit-maximizers, and the quantity where marginal revenue equals marginal cost maximizes profit.

   Marginal revenue is less than price because of the price effect. When a monopoly increases its quantity supplied, it has to drop the market price in order to sell more units. As a result, the marginal revenue from selling an additional unit is the price of that unit minus the decrease in revenue from receiving a lower price on all the units it was already selling.

4. (d) – If a monopoly's fixed costs increase, its marginal cost remains the same. That means it still has the same profit-maximizing quantity, so it will still charge the price corresponding to that quantity on the demand curve.

   Since the monopoly still produces the same quantity and sells it at the same price, its total revenue remains constant. Its total cost increase because of the increase in fixed costs. Profit equals total revenue minus total cost, so profit decreases.

5. (a) – See the plot below.



6. (b) – If there are consumers who can't get the good but who value it more than it costs to produce, then society is missing out on worthwhile trades.

7. (a) – Price discrimination is a firm's attempt to charge each consumer the highest price they can in order to maximize profits. That price is the consumer's willingness to pay.

8. (c) – When the monopoly charges a single price, all consumers with a willingness to pay higher than that price receive consumer surplus. When the monopoly perfectly price discriminates, all consumers pay their willingness to pay and receive zero consumer surplus.

9. (c) – Antitrust laws are meant to prevent companies from amassing market power and wielding it in a harmful way.

10. (c) – Natural monopolies form when firms have economies of scale. That's to say, natural monopolies form when firms have decreasing average total costs for the entire range of potential outputs. If average total costs are decreasing, that means marginal cost is less than average total cost. Profit equals $Q(P - ATC)$, so if natural monopolies set their price equal to marginal cost, their profits would be negative.

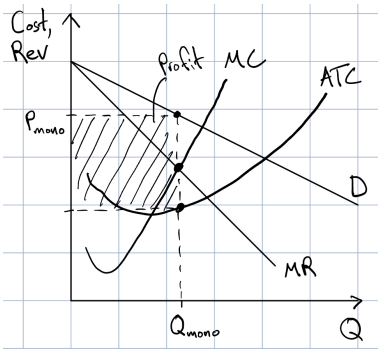
Questions for Review

1. Any pharmaceutical that has a patent is a government-created monopoly. On the one hand, the policy leads to higher prices for pharmaceuticals. On the other hand, it allows pharmaceutical companies to profit off of their research which incentivizes them to keep innovating better drugs. That's to say, the policy has its pros and cons.

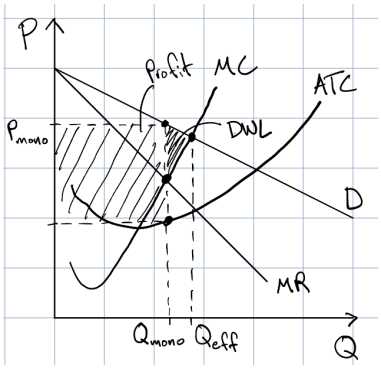
2. A natural monopoly is an industry in which one firm can produce the market quantity more cheaply than multiple firms could. In other words, it's an industry with decreasing average total cost over the range of feasible market quantities. The size of the market is important because the larger the market, the more likely it is for average total cost to increase. If average total cost increases, it might become profitable for additional firms to enter and make the market more competitive.

3. For a monopoly, marginal revenue is less than the price of its good because of the price effect. When a monopoly increases its production by a unit, total revenue ($P \cdot Q$) is pulled in opposite directions. On the one hand, quantity increases which pulls total revenue up. That's the output effect. On the other hand, the price of the good must decrease in order to sell an additional unit which pulls total revenue down. That's the price effect. Altogether, that means a monopoly's marginal revenue is the price of the additional unit it sells minus the decrease in revenue from dropping the price on all of the units it was already selling. If the price effect is greater than the output effect, a monopoly's marginal revenue can certainly be negative.

4.



5. A monopoly supplies a quantity at which marginal cost is less than the willingness to pay of the marginal consumer. That means supplying an additional unit would benefit the marginal consumer more than it would cost the monopoly. In other words, it would increase total surplus. Therefore, the monopoly provides an inefficiently low quantity.



6. Book publishers price discriminate by charging more for hardcover editions that come out initially and less for paperback editions that come out later. It allows them to charge people with a higher willingness to pay – big fans and those who want to stay on top of the zeitgeist – more and people with a lower willingness to pay – casual fans and people who have heard good things – less. Doing so increases their profit.

   Spotify price discriminates by charging college students less for a premium subscription. College students typically have less income and therefore a lower willingness to pay than adults with full-time jobs. By offering them a discount, Spotify can gain new customers who otherwise wouldn't be willing to pay for a subscription while still charging full price to its customers with a higher willingness to pay. Doing so increases Spotify's profit.

7. Antitrust laws give the government the power to regulate mergers between firms. A merger might be good if combining the two companies created synergies that allowed them to reduce the cost of production. A merger might be bad if it gave the combined firm the market power to charge an inefficiently high price.

8. First, charging the marginal cost would make the natural monopoly's profit negative. Natural monopolies occur in industries with economies of scale – industries that have decreasing average total cost. As we saw in chapter 14, if average total cost is decreasing, marginal cost must be lower than average total cost. Therefore, if regulators set the price equal to the marginal cost of production, the price would be below the average total cost.Profit $= Q(P - ATC)$, so profit would be negative, and the monopoly would exit the market.

   Second, if the monopoly knows it will always have to set its price equal to marginal cost, it has no incentive to reduce costs. If any reduction in cost comes with a reduction in price, the monopoly doesn't stand to benefit from cost cutting improvements. Accordingly, it won't prioritize them, and society might miss out on cost saving innovations.

Problems and Applications

1. (a)

$$\text{Total Revenue} = PQ$$
$$\text{Total Cost} = \$2 \text{ million} + \$10 \cdot Q$$
$$\text{Profit} = \text{Total Revenue} - \text{Total Cost}$$

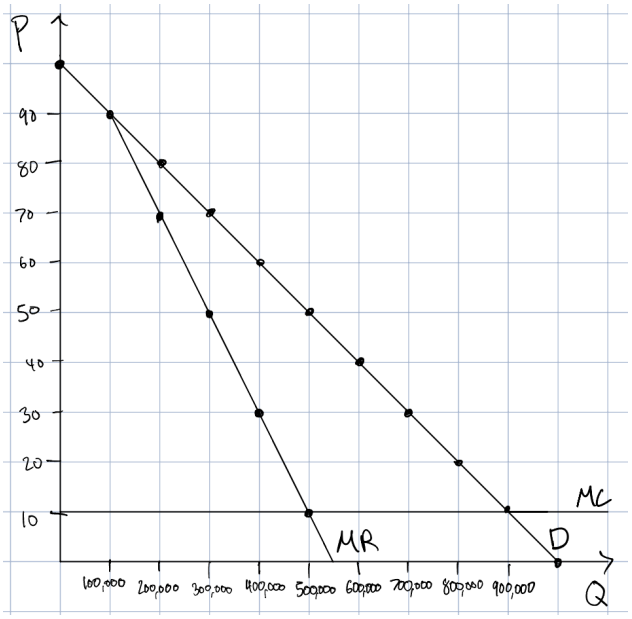
Price	Quantity Demanded	Total Revenue	Total Cost	Profit
$100	0 novels	$0	$2,000,000	-$2,000,000
90	100,000	9,000,000	3,000,000	6,000,000
80	200,000	16,000,000	4,000,000	12,000,000
70	300,000	21,000,000	5,000,000	16,000,000
60	400,000	24,000,000	6,000,000	18,000,000
50	500,000	25,000,000	7,000,000	18,000,000
40	600,000	24,000,000	8,000,000	16,000,000
30	700,000	21,000,000	9,000,000	12,000,000
20	800,000	16,000,000	10,000,000	6,000,000
10	900,000	9,000,000	11,000,000	-2,000,000
0	1,000,000	0	12,000,000	-12,000,000

A profit-maximizing publisher would produce 400,000 novels and charge a price of $60, or it would produce 500,000 novels and charge a price of $50. Both would maximize profit at $18,000,000.

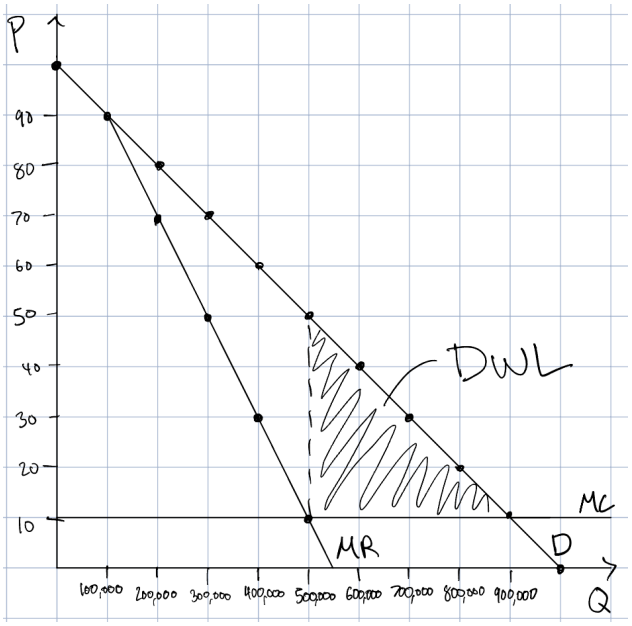
(b) Marginal revenue is less than price everywhere except the first non-zero quantity. That's because of the price effect. Every time a firm with market power wants to sell additional units, it has to lower its price to attract additional buyers. Charging a lower price will decrease the revenue it receives for all the units it was already selling. As a result, marginal revenue equals the price of the additional units the firm sells minus the decrease in revenue from cutting the price on all other units. Therefore, marginal revenue is less than the new price.

Price	Quantity Demanded	Total Revenue	Marginal Revenue
$100	0 novels	$0	–
90	100,000	9,000,000	90
80	200,000	16,000,000	70
70	300,000	21,000,000	50
60	400,000	24,000,000	30
50	500,000	25,000,000	10
40	600,000	24,000,000	-10
30	700,000	21,000,000	-30
20	800,000	16,000,000	-50
10	900,000	9,000,000	-70
0	1,000,000	0	- 90

(c) The marginal revenue and marginal cost curves cross at a quantity of 500,000. That signifies that 500,000 is the profit-maximizing quantity.



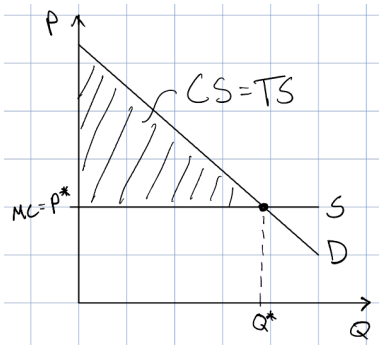
(d) The deadweight loss triangle is the surplus that society loses when the book publisher uses its market power to sell an inefficiently low quantity. The surplus-maximizing quantity is where the marginal cost curve and demand curve intersect – 900,000 novels. That's the quantity at which the cost of producing an additional unit is exactly equal to the benefit a consumer would get from that unit. When the book publisher sells a smaller quantity, society loses out on trades where the benefit exceeds the cost.



(e) It would not affect the publisher's decision. The publisher would still charge the price on the demand curve corresponding to the profit-maximizing quantity. The profit-maximizing quantity is the quantity where marginal cost equals marginal revenue. Since the author's compensation doesn't affect marginal cost or marginal revenue, nothing about the publisher's decision making process changes.

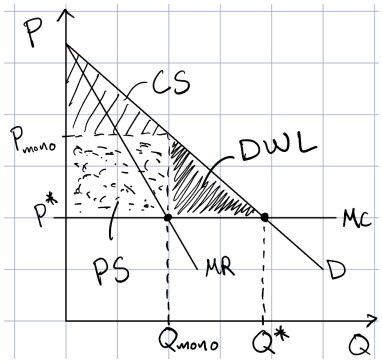
(f) The publisher would charge $10 because that's the price where the demand curve and marginal cost intersect. It would make a profit of -$2 million.

2. (a) Because all of the producers have constant marginal costs, all of the producers have horizontal individual supply curves. If all of the individual supply curves are horizontal, then market supply is also horizontal. That means producer surplus is zero, and consumer surplus equals total surplus.



(b) Relative to the competitive market, consumer surplus is smaller. Some of it transferred to producers, and some of it became deadweight loss. Total surplus is the combined areas of consumer surplus and producer surplus.

The deadweight loss is the result of the monopoly selling an inefficiently low quantity of groceries. There are some consumers who no longer purchase groceries even though the consumers value them more than they cost to produce.



3. (a)

$$\text{Total Revenue} = P \cdot Q$$

$$\text{Marginal Revenue} = \frac{\Delta TR}{\Delta Q}$$

Price	**Number of CDs**	**Total Revenue**	**Marginal Revenue**
\$24	10,000	\$240,000	–
22	20,000	440,000	\$20
20	30,000	600,000	16
18	40,000	720,000	12
16	50,000	800,000	8
14	60,000	840,000	4

(b) Selling 50,000 CDs at a price of \$16 would maximize profit. Since there are no fixed costs, average total cost equals average variable cost, \$5. That means

$$\begin{aligned}\text{Profit} &= Q(P - ATC) \\ &= 50,000(\$16 - \$5) \\ &= \$550,000\end{aligned}$$

(c) I'd advise her to ask for \$550,000. Her fee doesn't change the recording company's marginal cost, so it doesn't change the quantity they would produce. And if we assume that the given variable cost includes implicit cost, that's the price at which the recording company would be indifferent between producing her CD or not.

4. (a) The company will produce the quantity at which marginal cost equals marginal revenue. Marginal cost is \$0. Marginal revenue is:

Price	**Crossings (Thousands)**	**Total Revenue (Thousands)**	**Marginal revenue**
\$8	0	$\$8 \cdot 0 = \$0$	–
7	100	$7 \cdot 100 = 700$	$\frac{\$700-\$0}{100-0} = \$7$
6	200	$6 \cdot 200 = 1,200$	$\frac{1,200-700}{200-100} = 5$
5	300	$5 \cdot 300 = 1,500$	$\frac{1,500-1,200}{300-200} = 3$
4	400	$4 \cdot 400 = 1,600$	$\frac{1,600-1,500}{400-300} = 1$
3	500	$3 \cdot 500 = 1,500$	$\frac{1,500-1,600}{500-400} = -1$
2	600	$2 \cdot 600 = 1,200$	$\frac{1,200-1,500}{600-500} = -3$
1	700	$1 \cdot 700 = 700$	$\frac{700-1,200}{700-600} = -5$
0	800	$0 \cdot 800 = 0$	$\frac{0-700}{800-700} = -7$

The profit-maximizing quantity is the last quantity for which marginal revenue is greater than or equal to marginal cost, 400,000 crossings. The monopoly will charge the price corresponding to 400,000 crossings on the demand curve, \$4.

400,000 crossings is not the efficient level of output. At that quantity, the cost of providing another bridge crossing is less than the benefit a consumer would get from the crossing. The efficient quantity is the quantity at which the cost of providing another bridge crossing equals the benefit a consumer would get from the crossing. That's the intersection of marginal cost and demand. In this case, it's a quantity of 800,000 crossings.

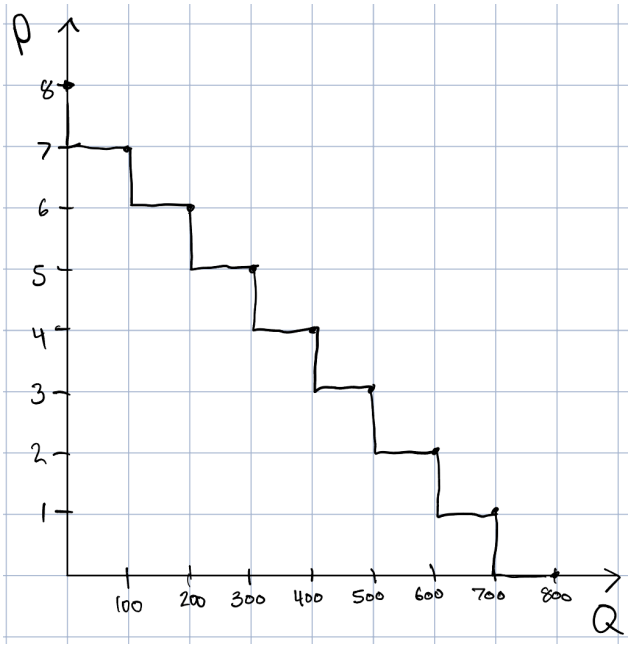
(b) It's profit would be

$$\begin{aligned}\text{Profit} &= TR - TC \\ &= 1,600,000 - 2,000,000 \\ &= -400,000\end{aligned}$$

Since profit is negative at the profit-maximizing quantity, the company should not build the bridge.

(c) If the government were to build the bridge, it would want to charge the price that led to the efficient number of crossings. That would be \$0, the marginal cost of providing a crossing. At that price, there would be 800,000 crossings.

(d) The demand curve looks approximately like this:



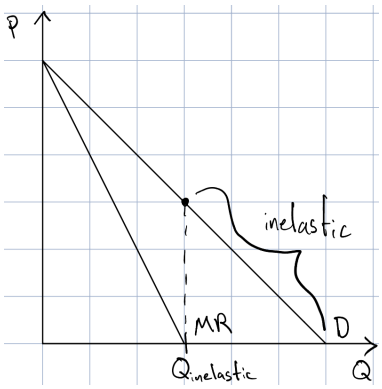
At a price of \$0, consumer surplus would be roughly:

$$\begin{aligned} CS &= \$7(100,000) + \$6(100,000) + \$5(100,000) + \$4(100,000) + \$3(100,000) + \$2(100,000) + \$1(100,000) \\ &= 100,000(\$7 + \$6 + \$5 + \$4 + \$3 + \$2 + \$1) \\ &= 100,000(28) \\ &= 2,800,000. \end{aligned}$$

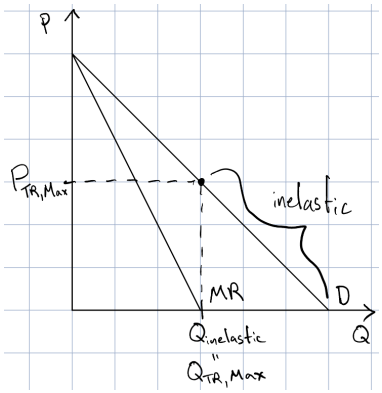
Therefore, the overall benefit from the bridge appears greater than the overall cost of \$2,000,000, so the government should build the bridge.

5. (a) At a quantity where the demand curve is inelastic, an increase in the price will cause a proportionately smaller decrease in quantity demanded. That means an increase in price will raise total revenue. At the same time, it will decrease total cost because the monopoly will have to supply less output to increase the price. Therefore, at a quantity where the demand curve is inelastic, an increase in price and the corresponding decrease in quantity would increase profit. That means a profit-maximizing monopoly would never produce a quantity at which the demand curve is inelastic.

   (b) The demand curve is inelastic everywhere that an increase in quantity leads to a proportionately larger decrease in price. That means the demand curve is inelastic everywhere that an increase in quantity causes total revenue to decrease. Total revenue is decreasing everywhere that marginal revenue is negative.



(c) Total revenue is increasing everywhere that marginal revenue is positive and decreasing everywhere that marginal revenue is negative. That means the total revenue maximizing quantity is the quantity where marginal revenue switches from positive to negative.



6. (a) In this scenario, there is no variable cost, so total cost is constant and equals the fixed cost, \$2,000. Therefore, in order to maximize profits, all we need to do is maximize total revenue. For adults, the price that maximizes total revenue is \$7, and for kids, the price that maximizes total revenue is \$4. At those two prices,

$$\begin{aligned}\text{Profit} &= \$7(300) + \$4(200) - \$2,000 \\ &= \$900\end{aligned}$$

(b) Since total cost is constant, I still just need to maximize total revenue. Looking at the table, the prices that could feasibly do so are \$7, \$5, and \$4.

Price	**n Tickets**	**Total Revenue**
\$7	$300 + 0$	$\$7 \cdot 300 = \$2{,}100$
5	$300 + 100$	$5 \cdot 400 = 2{,}000$
4	$300 + 200$	$4 \cdot 500 = 2{,}000$

A price of \$7 maximizes total revenue. No theater for the youth. I make a profit of \$2,100 − \$2,000 = \$100.

(c) I am worse off by \$800 because my profit decreased from \$900 to \$100.

The adults are no better or worse off.

100 of the children are worse off by at least \$1 and at most \$2 each. They have a willingness to pay between \$5 and \$6, and they paid a price of \$4 before the law. That means they each lost consumer surplus of \$1 to \$2.

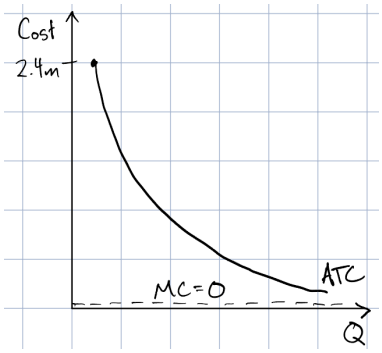
The other 100 children are worse off by at most \$1 each. They have a willingness to pay between \$4 and \$5 and paid a price of \$4 before the law. That means they each lost consumer surplus of \$1 or less.

(d) In part (a), I would still charge the same prices because I still just want to maximize total revenue, but my profit would decrease from \$900 to \$400.

In part (b), I wouldn't put on the play because it would be impossible for me to make positive profit. My profit would be \$0.

In part (c), I would be worse off by \$900 because my profit would go to zero. 100 adults would be worse off by at least \$2 and at most \$3 each because they have a willingness to pay between \$9 and \$10, and they only paid \$7. 100 adults would be worse off by at least \$1 and at most \$2 because they have a willingness to pay between \$8 and \$9, and they only paid \$7. The last 100 adults would be worse off by at most \$1 because they have a willingness to pay between \$7 and \$8, and they only paid \$7. The children are just as worse off as they were before.

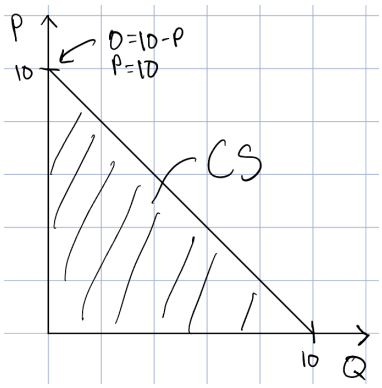
7. (a) The museum is a natural monopoly because it has decreasing average total cost for all quantities of output.



(b) If it were free for each person to visit, then each person's quantity demanded would be

$$\begin{aligned} Q_D &= 10 - 0 \\ &= 10. \end{aligned}$$

Consumer surplus would be



$$\begin{aligned} CS &= \frac{1}{2}(10)(10) \\ &= 50. \end{aligned}$$

Therefore, each consumer would get \$50 − \$24 = \$26 worth of benefit.

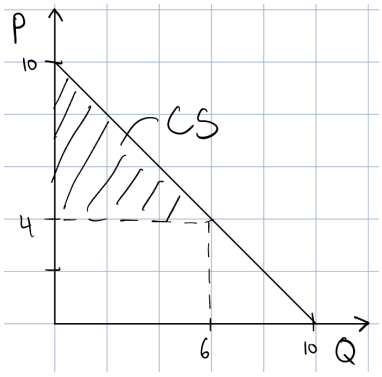
(c)

$$\begin{aligned} \text{Visits per Person} &= 10 - P \\ \text{Total Revenue} &= \text{Visits per Person} \cdot 100{,}000 \cdot P \\ \text{Profit} &= \text{Total Revenue} - \$2.4 \text{ million} \end{aligned}$$

Price	**Visits per Person**	**Total Revenue**	**Profit**
\$2	8	\$1.6 million	-\$0.8 million
3	7	2.1 million	-0.3 million
4	6	2.4 million	0 million
5	5	2.5 million	-0.1 million

The lowest price the museum can charge without incurring losses is \$4.

(d)



$$\begin{aligned} CS &= \frac{1}{2}(10-4)(6) \\ &= 18 \end{aligned}$$

Everyone is worse off with the admission fee because all consumers receive less consumer surplus.

(e) In the absence of an admission fee, the museum might get over crowded. At that point, it would become rival in consumption because one person's visit would detract from another's. In other words, visits would have a negative externality. An admission fee would internalize that externality.

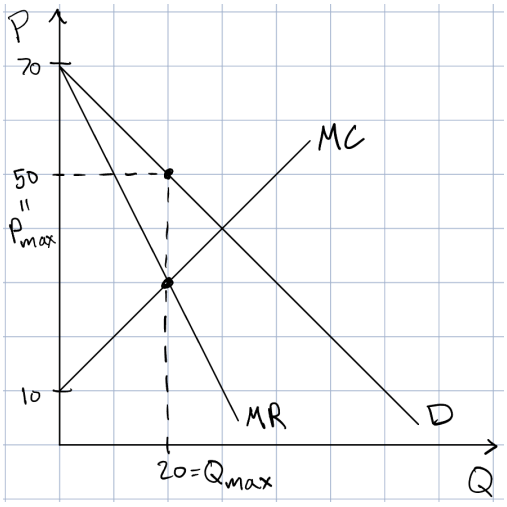
A less technical problem – taxes are unpopular, so an admission fee would be politically easier.

8. (a) Quantity:

$$\begin{aligned} 70 - 2Q &= 10 + Q \\ 60 &= 3Q \\ 20 &= Q_{max} \end{aligned}$$

Price:

$$\begin{aligned} P_{max} &= 70 - Q_{max} \\ &= 70 - 20 \\ &= 50 \end{aligned}$$



(b) Price:

$$P_{ceiling} = 0.9(50) = 45$$

Quantity demanded:

$$45 = 70 - Q_D$$
$$Q_D = 25$$

To figure out how much Mr. Potter produces, we have to consider how his marginal revenue changes with the price ceiling. For the first 25 units that he produces, his marginal revenue would be constant at \$45. The marginal cost of the 25th unit is only \$10 + \$25 = \$35, so it increases Mr. Potter's profit to produce 25 units.

For the 26th unit, Mr. Potter could charge $P = \$70 - \$26 = \$44$. Total revenue would be $TR = \$44 \cdot 26 = \$1{,}144$. Total revenue from 25 units would be $TR = \$45 \cdot 25 = \$1{,}125$. That means marginal revenue for the 26th unit would be $MR = \$1{,}144 - \$1{,}125 = \$19$. Marginal cost would be $MC = \$10 + \$26 = \$36$. Since marginal cost is greater than marginal revenue, Mr. Potter would not produce a 26th unit. Therefore, Mr. Potter does produce 25 units, the quantity demanded.

(c) Uncle Billy is not right in this case. As we showed above, Mr. Potter would supply exactly the quantity demanded. In general, price ceilings only cause shortages when the ceiling is below the efficient price – the price where marginal cost equals marginal willingness to pay.

(d) Price:

$$P_{ceiling} = 0.5(50) = 25$$

Quantity demanded:

$$25 = 70 - Q_D$$
$$Q_D = 45$$

Quantity supplied:

In this scenario, Mr. Potter's marginal revenue is \$25 for the first 45 units. Marginal cost equals \$25 when Mr. Potter supplies 15 units. Therefore, Mr. Potter will only supply 15 units.

Shortage:

Uncle Billy is right in this case. Quantity demanded is 45, and quantity supplied is 15, so the price ceiling creates a 30 unit shortage.

10. (a)

$$\begin{aligned} MC &= MR \\ 100 + 10Q &= 1{,}000 - 20Q \\ 30Q &= 900 \\ Q_{max} &= 30 \\ P_{max} &= 1{,}000 - 10(30) \\ &= 700 \end{aligned}$$

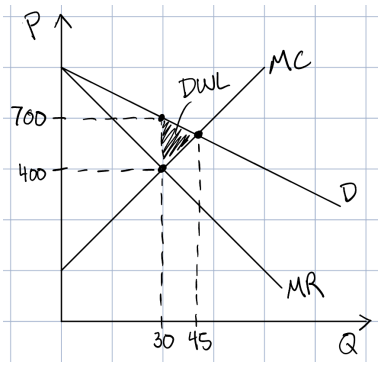
(b)

$$\begin{aligned} MC &= WTP \\ 100 + 10Q &= 1{,}000 - 10Q \\ 20Q &= 900 \\ Q_{eff} &= 45 \end{aligned}$$

(c) Price at which $MC = MR$:

$$P = 1{,}000 - 20Q_{max} = 1{,}000 - 20(30) = 400$$

Diagram:



Deadweight loss:

$$DWL = \frac{1}{2}(700 - 400)(45 - 30) = 2{,}250$$

(d) i. Marginal cost doesn't change, so neither do the profit-maximizing price and quantity. That means deadweight loss stays the same as well.

ii. Again, marginal cost doesn't change, so the profit-maximizing price and quantity don't change, and neither does the deadweight loss. Another way to look at it – the company still wants to make profit as large as possible before the 50% payout, so they still operate the same way they would without the payout.

iii. Now, the firm's marginal cost is given by

$$MC = 100 + 10Q + 150 = 250 + 10Q.$$

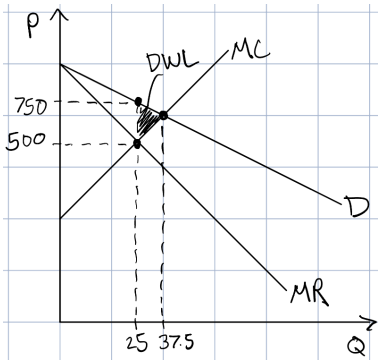
Therefore, the profit-maximizing quantity is

$$\begin{aligned} MC &= MR \\ 250 + 10Q &= 1{,}000 - 20Q \\ 30Q &= 750 \\ Q_{max} &= 25, \end{aligned}$$

and the profit-maximizing price is

$$P_{max} = 1{,}000 - 10(25) = 750.$$

The diagram now looks like this:



where the efficient quantity is

$$\begin{aligned} MC &= WTP \\ 250 + 10Q &= 1{,}000 - 10Q \\ 20Q &= 750 \\ Q_{eff} &= 37.5, \end{aligned}$$

and the price where marginal cost equals marginal revenue is

$$P = 250 + 10(25) = 500$$

Therefore, deadweight loss is

$$DWL = \frac{1}{2}(750 - 500)(37.5 - 25) = 1{,}562.5$$

iv. The firm's new marginal revenue function is

$$MR = 0.5(1{,}000 - 20Q) = 500 - 10Q.$$

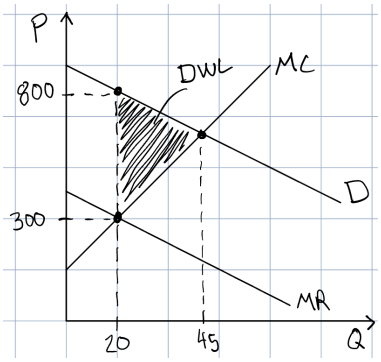
As a result, the profit-maximizing quantity is

$$\begin{aligned} MC &= MR \\ 100 + 10Q &= 500 - 10Q \\ 20Q &= 400 \\ Q_{max} &= 20, \end{aligned}$$

and the profit-maximizing price is

$$P_{max} = 1{,}00 - 10(20) = 800.$$

The new diagram looks like



where the price at which marginal cost and marginal revenue intersect is

$$MR = 500 - 10(20) = 300.$$

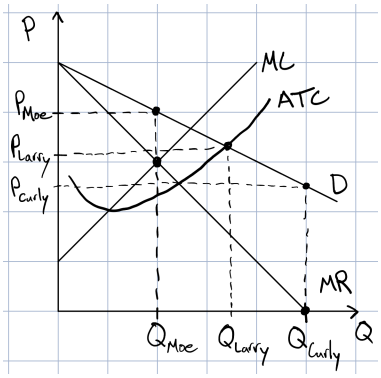
Therefore, deadweight loss is

$$DWL = \frac{1}{2}(800 - 300)(45 - 20) = 6{,}250$$

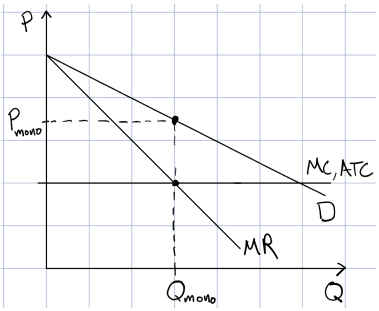
11. Moe is the easiest. Moe wants to maximize profit, so Moe wants to produce the quantity where marginal cost equals marginal revenue. He then wants to charge the maximum willingness to pay for that quantity which is given by the height of the demand curve.

    Curly wants to bring in as much revenue as possible, so Curly wants to sell every drink with positive marginal revenue. That means Curly wants to sell the quantity where marginal revenue equals zero. He then wants to charge the maximum willingness to pay for that quantity which is given by the height of the demand curve.

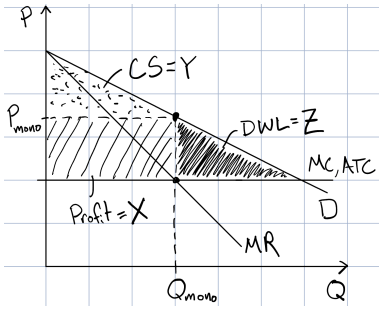
    Larry wants to sell as many drinks as possible without losing money, so Larry wants to produce the highest quantity at which profit equals zero. Profit equals zero when price equals average total cost. Therefore, Larry should produce the quantity where average total cost and demand intersect because that's the greatest quantity at which the price people are willing to pay equals average total cost.



12. (a)



(b)



(c) The monopolist's profit is $X + Y + Z$ because it produces the quantity at which marginal cost equals willingness to pay, and it charges each consumer their willingness to pay.

(d) The monopolist's profit increases by $Y + Z$. Total surplus increases by Z. The change in profit is greater because the monopolist captures both the deadweight loss and the consumer surplus as profit.

(e) If $Y + Z > C$, then the benefits of price discrimination outweigh the costs for the monopolist. In words, if the increase in the monopolist's profit is greater than the cost, then the monopolist will price discriminate

(f) If $Z > C$, then the benefits of price discrimination outweigh the costs for society. In words, if the increase in total surplus is greater than the cost, then the social planner will have the monopolist price discriminate.

(g) The monopolist has more of an incentive to price discriminate than the social planner because the monopolist benefits more from price discrimination than society does. If $Y + Z > C > Z$, then the monopolist will price discriminate even though doing so is not socially desirable. In words, if the increase in the monopolist's profit is greater than the cost but the cost is greater than the increase in total surplus, the monopolist will price discriminate even though doing so is not socialy desirable.

Non-Textbook Questions

1. (a)

$$\begin{aligned} Q &= 300 - P \\ P &= 300 - Q \end{aligned}$$

(b)

$$\begin{aligned} TR &= P \cdot Q \\ &= (300 - Q) \cdot Q \\ &= 300Q - Q^2 \end{aligned}$$

(c)

$$\begin{aligned} MR &= \frac{\mathrm{d}TR}{\mathrm{d}Q} \\ &= \frac{\mathrm{d}}{\mathrm{d}Q}(300Q - Q^2) \\ &= 300 - 2Q \end{aligned}$$

(d)

$$\begin{aligned} MC &= \frac{\mathrm{d}TC}{\mathrm{d}Q} \\ &= \frac{\mathrm{d}}{\mathrm{d}Q}(1{,}200 + 0.5Q^2) \\ &= Q \end{aligned}$$

(e) Quantity:

$$\begin{aligned} MC &= MR \\ Q &= 300 - 2Q \\ 3Q &= 300 \\ Q_{mono} &= 100 \end{aligned}$$

Price:

$$P_{mono} = 300 - Q = 300 - 100 = 200$$